

On the Convergence Rate of the Least Trimmed Squares Algorithm

Samit Ghosh¹, Michael Pokojovy^{*1}, and Guohui Song¹

¹Old Dominion University, Department of Mathematics and Statistics, 4300 Elkhorn Ave,
Norfolk, VA 23529

November 3, 2025

Abstract

The least trimmed squares (LTS) approach is a popular alternative to ordinary least squares (OLS) commonly used to robustly fit linear regression models in the presence of outliers or other model violations. While a closed-form solution exists for the quadratic optimization problem underlying the OLS estimator, the mixed integer-continuous optimization problem associated with the LTS estimator is typically solved by iteratively applying the so-called concentration or C-step to a set of properly selected warm starts. Although the C-step is known to monotonically reduce the objective function leading to convergence in a finite number of steps, no error estimation mechanisms presently exist in the literature. By studying an equivalent formulation of the problem over a convex domain, we prove that the conventional LTS iteration is a special case of the well-known Frank-Wolfe gradient descent method. This furnishes an $\mathcal{O}(1/t)$ convergence of the LTS algorithm to a local minimum of the trimmed sum of squares objective function with respect to a suitable Bregman divergence and provides a convenient error quantification tool.

Keywords: Robust estimation, Frank-Wolfe gradient descent, concentration step, error control, Bregman divergence.

1 Introduction

Regression analysis is widely used in supervised learning to analyze statistical relationship between a dependent response variable and one or multiple independent predictor variables. Linear regression specifically focuses on linear models and offer multiple advantages such as simplicity, interpretability and computationally efficient implementation via ordinary least squares (OLS). Application fields include economics, business, finance, engineering, biomedical and health sciences, etc. However, despite widespread use, the classical OLS estimator is highly sensitive to even small departures from the postulated model assumptions, particularly potential presence of outliers. In real-world datasets, such deviations are very common and can severely distort the estimates produced by OLS, leading to poor predictive performance and misleading conclusions (Boyd and Mattingley, 2007).

The classical multiple linear regression model can be expressed as:

$$y_i = \boldsymbol{\beta}^\top \mathbf{x}_i + \varepsilon_i, \tag{1.1} \boxed{\text{eq:LR}}$$

where $\mathbf{x}_i \in \mathbb{R}^p$ is the vector of explanatory variables (possibly including an intercept term), $\boldsymbol{\beta} \in \mathbb{R}^p$ is the coefficient vector and $\varepsilon_i \stackrel{\text{i.i.d}}{\sim} \mathcal{N}(0, \sigma^2)$ is the error term. Given n observations pairs $(\mathbf{x}_i, y_i), i =$

^{*}Corresponding author. Email: mpokojovy@odu.edu

$1, \dots, n$, we are interested in estimating β . Assuming design matrix $\mathbf{X}$ has the full column rank of p , the OLS estimator is given as the unique minimizer of the residual sum of squares (RSS)

$$\hat{\beta}_{\text{OLS}} = \arg \min_{\beta \in \mathbb{R}^p} \sum_{i=1}^n \left(y_i - \beta^\top \mathbf{x}_i \right)^2. \quad (1.2) \quad \boxed{\text{eq:OLS}}$$

hat can be explicitly expressed as

$$\hat{\beta}_{\text{OLS}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (1.3) \quad \boxed{\text{eq:cls}}$$

Being closely connected with the maximum likelihood estimation (MLE), this approach works well if the dataset is sampled from the underlying true model. Despite attractive statistical properties under the Gaussianity assumptions in Equation (1.1), the OLS estimator performs poorly in the presence of outliers, which can significantly bias parameter estimates due to the unbounded nature of the quadratic loss function. Even a single outlying observation can exert excessive influence on the fit, leading to inaccurate predictions and misleading inference. This issue becomes even more critical in higher dimensions or for heavy-tailed error distributions. One of the most widely adopted frameworks to model the presence of outliers is the Huber contamination model (Huber, 1992):

$$\mathcal{F}_\epsilon = (1 - \epsilon)\mathcal{G} + \epsilon\mathcal{H}$$

for some contamination proportion $\epsilon \in [0, 1/2)$ where $\mathcal{G}$ represents the uncontaminated distribution and $\mathcal{H}$ denotes an arbitrary contaminating distribution. Typically, both $\mathcal{G}$ and $\mathcal{H}$ are unknown. Even a small value of ϵ can cause substantial distortion in OLS estimates (Hadi and Luceño, 1997). To overcome this challenge, in addition to the Least Trimmed Squares (LTS) (Rousseeuw, 1985) investigated in this paper, a variety of robust statistical methodologies have been proposed in the literature including Least Median of Squares (LMS) (Rousseeuw, 1984), Maximum Trimmed Likelihood estimators (MTLE) (Tatsuoka and Tyler, 1988), S-estimators (Rousseeuw and Yohai, 1984), MM-estimators (Yohai, 1987), etc. See Rousseeuw and Leroy (2005) for a detailed description of these and other classical methods. Cellwise contamination models have also recently attracted considerable attention (Filzmoser et al., 2020).

The breakdown point (Donoho and Huber, 1983) is an important gauge of an estimator’s quantitative robustness. It is typically defined as the smallest portion of a contamination of sample (with size n) for which the estimator becomes uninformative.

While traditional MLE and MLS have the breakdown point of $\frac{1}{n}$ and asymptotic breakdown points of 0, high-breakdown estimators like properly tuned LTS or MTLE attain the maximum asymptotic breakdown point of 0.5. When matched for the breakdown point, statistical estimators differ in terms of their statistical efficiency, however, reweighted versions of respective “raw” estimators that are known to improve the efficiency.

Like most robust estimators, the LTS estimator is connected with significant computational challenges due to the combinatorial nature of the underlying optimization problem. Specifically, it requires selecting the best subset of $h \geq \lfloor (n + p + 1)/2 \rfloor$ observations from n to minimize the RSS objective, which involves evaluation of up to $\binom{n}{h}$ possible subsets. This exhaustive search becomes computationally infeasible even for moderately sized datasets. In fact, computing the exact LTS solution is classified as an NP-hard problem (Davies and Gather, 1993; Stromberg, 1993). One reason behind the popularity of the LTS estimator is the availability of computational heuristics.

Notable methodologies include the Fast-LTS algorithm (Rousseeuw and Driessen, 1999), branch-and-bound (Boyd and Mattingley, 2007), heuristic meta-algorithms like genetic algorithms and simulated annealing have also been explored (Maronna et al., 2006), etc.

Being among the most popular computational techniques, the Fast-LTS algorithm (Rousseeuw and Driessen, 1999) combines subsampling with iterative application of a concentration (or C-)step. Its implementation has been widely adopted in statistical software, increasing its accessibility to practitioners across fields.

This method starts from a selected subset of data, iteratively updating the regression fit and subset to minimize the trimmed residual sum of squares. Each C-step guarantees a monotonic decrease in the objective function and convergence to a stationary point in a finite number of steps.

Unlike OLS, for which a closed-form solution (viz. Equation (1.3)) is known, computing the LTS estimator involves solving an NP-hard optimization problem. In this paper, we start with the former mixed integer optimization problem and introduce an equivalent continuous relaxation. This perspective reveals that the Fast-LTS algorithm as a special case of the Frank-Wolfe (FW) gradient descent algorithm – a classical first-order method for constrained optimization over convex domains.

By establishing the concavity of the objective function, we further use the recent result of Yurtsever and Sra (2022) to obtain an $\mathcal{O}(1/t)$ convergence in the iteration count t with respect to the usual Frank-Wolfe gap – a special case of the Bregman divergence. In addition to proving that stationary points of the Fast-LTS algorithm are local minimizers of the LTS objective, this also allows us to develop an error estimation mechanism that quantifies the progress of the algorithm at each step—something previously unavailable in the LTS literature.

The following parts of the article is organized like the procedure and formulation of the FAST-LTS Algorithm in Section 2. In Section 3, we establish a mixed-integer continuous optimization problem where we applied F-W. The implementation of FW is yet surprising discovery in our work as an alternative of C-step algorithm (Proposition 3.4) which formulated in Section 2 and is discussed in Section 3. Our mathematical support for the above claim gets the foundation by the analysis done in Section 4 that shows the insight of convergence result. We did a simulation study and their result is described in Section 5. Finally, the article concludes with some closing thoughts in Section 6 with a summary and discussion that situates our findings and the broader context of robust regression and optimization.

2 The Fast-LTS Algorithm

(sec:lts) We will introduce the LTS estimator and briefly protocol the Fast-LTS estimator of Rousseeuw and Driessen (2006) along with the underlying C-step.

To this end, consider n observations pairs $(\mathbf{x}_i, y_i) \in \mathbb{R}^p \times \mathbb{R}, i = 1, \dots, n$. Assuming the multiple regression model in Equation (1.1), the residuals for any given parameter vector $\beta \in \mathbb{R}^p$ are defined as

$$r_i(\beta) = (y_i - \beta^\top \mathbf{x}_i)^2 \quad \text{for } i = 1, \dots, n.$$

The LTS estimator is a robust alternative to the OLS estimator that minimizes the sum of squared residuals after trimming a certain number of large residuals. Specifically, for a given integer $\lfloor (n+p+1)/2 \leq h \leq n$ (Rousseeuw and Leroy, 2005), the LTS estimator finds the parameter vector β that minimizes the sum of h smallest squared residuals, i.e.,

$$\hat{\beta}_{\text{LTS}} = \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \min_{\mathbf{w} \in \mathcal{M}_0} L(\mathbf{w}, \beta) \quad \text{with} \quad L(\mathbf{w}, \beta) := \sum_{i=1}^n w_i r_i(\beta), \quad (2.1) \quad \boxed{\text{eq:lts}}$$

where $\mathcal{M}_0 = \{\mathbf{w} \in \{0, 1\}^n \mid \|\mathbf{w}\|_1 = h\}$ is the set of all binary vectors of length n with exactly h non-zero entries, and $\|\cdot\|_1$ denotes the l_1 norm. The former binary encoding is used to specify which residuals are included (weight 1) and which residuals are excluded (weight 0) from the trimmed sum.

The Fast-LTS algorithm [Rousseeuw and Driessen \(2006\)](#) iteratively applies the C-step to solve the optimization problem in Equation (2.1). For a given value of β at current iteration, the algorithm finds the $\mathbf{w} \in \mathcal{M}_0$ that minimizes $L(\mathbf{w}, \beta)$. Then, it updates β by minimizing $L(\mathbf{w}, \beta)$ for the selected $\mathbf{w}$. The update step can be expressed as follows: (See [Algorithm 1](#))

Algorithm 1 C-step update

<alg:cstep>

- 1: Let $\beta_C^{(0)} \in \mathbb{R}^p$ and trimming parameter h .
- 2: **for** $k = 0, 1, 2, \dots$, **do**
- 3: Compute $r_i^2 := (y_i - \beta_C^{(k)\top} \mathbf{x}_i)^2$
- 4: Compute $\mathbf{w}_C^{(k+1)} := \arg \min_{\mathbf{w} \in \mathcal{M}_0} L(\mathbf{w}, \beta_C^{(k)})$
- 5: i.e, set $\mathbf{w}_C^{(k+1)} = 1$ for the h smallest r_i^2
- 6: Compute $\beta_C^{(k+1)} := \arg \min_{\beta \in \mathbb{R}^p} L(\mathbf{w}_C^{(k+1)}, \beta)$
- 7: **if** $L(\mathbf{w}_C^{(k+1)}, \beta_C^{(k+1)}) \leq L(\mathbf{w}_C^{(k)}, \beta_C^{(k)})$ **and**
- 8:
- 9: $\mathbf{w}_C^{(k+1)} = \mathbf{w}_C^{(k)}$ **then**
- 10: **return** $\mathbf{w}_C^{(k+1)}$
- 11: **end if**
- 12: **end for**
- 13: **return** $\beta_C^{(k+1)}$

Starting from an initial estimate $\beta_C^{(0)}$, for $k = 0, 1, 2, \dots$,

$$\begin{cases} \mathbf{w}_C^{(k+1)} = \arg \min_{\mathbf{w} \in \mathcal{M}_0} L(\mathbf{w}, \beta_C^{(k)}), \\ \beta_C^{(k+1)} = \arg \min_{\beta \in \mathbb{R}^p} L(\mathbf{w}_C^{(k+1)}, \beta), \end{cases} \quad (2.2) \quad \boxed{\text{eq:iterates}}$$

where the trimmed SSE objective L is defined in Equation (2.1). In the $\mathbf{w}$ -update perspective, the C-step is explicitly an optimal swap. It takes the current subset defined by $\mathbf{w}_C^{(k)}$, fits a LTS model, and then optimally re-selects the h best points according to this new model to form $\mathbf{w}_C^{(k+1)}$. The [Theorem 2.1](#) proves that this specific swap can never worsen the objective.

Let the residuals at iteration k be $r_i^{(k)} = \{r_i(\beta_C^{(k)}) \mid i = 1, \dots, n\}$ then corresponding order statistics of the residuals is denoted by ([Rousseeuw and Driessen, 2006](#))

$$r_{(1)}^{(k)} \leq r_{(2)}^{(k)} \leq \dots \leq r_{(n)}^{(k)}. \quad (2.3) \quad \boxed{\text{eq:order_res}}$$

Now, let $\{i_{(1)}, i_{(2)}, \dots, i_{(n)}\}$ be the corresponding observation indices such that

$$r_{(j)}^{(k)} = r_{i_{(j)}}^{(k)}, \quad j = 1, \dots, h. \quad (2.4) \quad \boxed{\text{eq:residue_ind}}$$

Then, the optimization problem in Equation (2.2)—a linear integer program—has the explicit solution in $I^{(k)} = \{i_{(1)}, i_{(2)}, \dots, i_{(h)}\}$, that is solved by sorting the residuals in Equation (2.3), and

selecting the h smallest ones for Equation (2.4) forming a index set denoted by $I^{(k)}$ (Rousseeuw and Driessen, 2006). Ties are broken arbitrarily when they occur. Here in the following, the optimal binary weights are in the Equation (2.1) then expressed as

$$[\mathbf{w}_C^{(k+1)}]_j = \begin{cases} 1, & \text{if } j \in I^{(k)}, \\ 0, & \text{otherwise.} \end{cases}$$

By the definition of optimality:

$$\sum_{i=1}^n [\mathbf{w}_C^{(k+1)}]_i \cdot r_i^2(\boldsymbol{\beta}_C^{(k)}) \leq \sum_{i=1}^n [\mathbf{w}_C^{(k)}]_i \cdot r_i^2(\boldsymbol{\beta}_C^{(k)}). \quad (2.5) \quad \text{eq:inequality_}$$

Subsequently, the updated parameter vector $\boldsymbol{\beta}_C^{(k+1)}$ is the global minimizer of the least squares objective on the new subset:

$$\boldsymbol{\beta}_C^{(k+1)} = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^p} \sum_{i \in I^{(k+1)}} r_i^2(\boldsymbol{\beta}). \quad (2.6) \quad \text{eq:unique_beta}$$

It, therefore, trivially follows that:

$$\sum_{i=1}^n [\mathbf{w}_C^{(k+1)}]_i r_i^2(\boldsymbol{\beta}_C^{(k+1)}) \leq \sum_{i=1}^n [\mathbf{w}_C^{(k+1)}]_i r_i^2(\boldsymbol{\beta}_C^{(k)}). \quad (2.7) \quad \text{eq:inequality_}$$

Combining inequalities (2.5) and (2.7) yields the final result:

$$\sum_{i=1}^n [\mathbf{w}_C^{(k+1)}]_i r_i^2(\boldsymbol{\beta}_C^{(k+1)}) \leq \sum_{i=1}^n [\mathbf{w}_C^{(k)}]_i r_i^2(\boldsymbol{\beta}_C^{(k)}). \quad (2.8) \quad \text{eq:inequality}$$

Equality holds for the inequality (2.8) only if both (2.5) and (2.7) are equal, which occurs if and only if the subset of indices remains unchanged as $I^{(k)} = I^{(k+1)}$ and

$$\mathbf{w}_C^{(k+1)} = \mathbf{w}_C^{(k)} \quad (2.9) \quad \text{eq:weight_equal}$$

also the least squares solution $\boldsymbol{\beta}_C^{(k+1)}$ on that subset $I^{(k+1)}$ is identical to the previous estimate $\boldsymbol{\beta}_C^{(k)}$. The $\boldsymbol{\beta}$ -update step is the OLS problem (viz. (1.2)). Specifically, the objective function

$$L(\mathbf{w}_C^{(k+1)}, \boldsymbol{\beta}) = \sum_{i=1}^n [\mathbf{w}_C^{(k+1)}]_i (y_i - \boldsymbol{\beta}^\top \mathbf{x}_i)^2. \quad (2.10) \quad \text{eq:obj.reduced}$$

is a quadratic function of $\boldsymbol{\beta}$ for any fixed $\mathbf{w}_C^{(k+1)}$. The unique minimizer $\boldsymbol{\beta}_C^{(k+1)}$ in Equation (2.6) can be obtained from Fermat's optimality conditions by setting the gradient

$$\nabla_{\boldsymbol{\beta}} L(\mathbf{w}_C^{(k+1)}, \boldsymbol{\beta}_C^{(k+1)}) = 0.$$

furnishing

$$\boldsymbol{\beta}_C^{(k+1)} = \left(\sum_{i \in I^{(k)}} \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\sum_{i \in I^{(k)}} y_i \mathbf{x}_i \right) \quad (2.11) \quad \text{eq:estimator}$$

on the strength of **Proposition A.1**. Describing the **Theorem 2.1** in terms of elegantly captures the iterative process as a sequence of discrete weight assignments that monotonically improve

the LTS criterion. The condition for convergence (a stationary point) becomes the simple and intuitive fixed-point condition in Equation (2.9). The classical selection of set $\mathbf{w}_C^{(k)}$ in C-step that confirms the sum of residuals is monotonically decreasing over $\mathbf{w} \in \mathcal{M}_0$. Though C-step guarantees monotonic decrease (or stationarity) of the objective, it does not ensure convergence to the global minimum. So the trimmed sum is minimum for the indexed set $I^{(k)}$ with estimator (2.11). This result was first proved for the Least Median of Squares (LMS) estimator by (Rousseeuw, 1984), and subsequently generalized to the Least Trimmed Squares (LTS) estimator by (Rousseeuw and Leroy, 2005). The algorithm may terminate at a stationary point, which could be a strict local minimum or, in degenerate cases, a saddle-like configuration.

$\langle \text{thm:cstep} \rangle$

Theorem 2.1. *Let $\mathbf{w}_C^{(k)} \in \{0, 1\}^n$ be a binary weight vector with $\sum_{i=1}^n [\mathbf{w}_C]_i^{(k)} = h$, and associated with the subset $I^{(k)} = \{i : [\mathbf{w}_C^{(k)}]_i = 1\}$ and $\beta_C^{(k)}$ be the least squares estimate computed using the observations with weights $\mathbf{w}_C^{(k)}$. Then, the C-step update calculates a new weight vector $\mathbf{w}_C^{(k+1)}$ over index set $I^{(k+1)}$ with same cardinality and satisfies Equation (2.8) and equality holds if and only if Equation (2.9) is true. Hence, the trimmed residual sum of squares is non-increasing across iterations.*

3 The Frank-Wolfe Algorithm

$\langle \text{sec:fw} \rangle$

Frank and Wolfe (1956) originally introduced the FW algorithm for minimizing convex quadratic functions over polyhedral sets and established an $\mathcal{O}(1/t)$ convergence rate. This result was later generalized by Levitin and Polyak (1966), who proved that the same convergence rate holds for any smooth convex function defined over a compact convex domain. Canon and Cullum (1968) further demonstrated that this rate is optimal. The analysis of FW in non-convex settings was presented by Lacoste-Julien (2016), who showed that the algorithm converges to a stationary point with a gap function decreasing at rate $\mathcal{O}(1/\sqrt{t})$. This rate improves to $\mathcal{O}(1/t)$ for minimization of concave functions Yurtsever and Sra (2022). Luss and Teboulle discussed an example of concave minimization via the FW algorithm in (Luss and Teboulle, 2013).

In this section, we will reformulate the LTS estimator in Equation (2.1) as a constrained optimization problem in $\mathbf{w}$ and then apply the FW algorithm to solve it. Specifically, we first switch the minimization order in Equation (2.1). For each $\mathbf{w} \in [0, 1]^n$ with $\sum_{i=1}^n w_i \geq h$, let

$$F(\mathbf{w}) := L(\mathbf{w}, \beta^{\mathbf{w}}) = \sum_{i=1}^n w_i r_i(\beta^{\mathbf{w}}) \quad (3.1) \quad \boxed{\text{eq:f_vw}}$$

where $\beta^{\mathbf{w}} = \arg \min_{\beta \in \mathbb{R}^p} L(\mathbf{w}, \beta)$ and the minimum exists uniquely under the conditions of **Proposition A.1**. The minimization of F over the constraint set $\mathcal{M}_0$ can be solved uniquely if the constraint set $\mathcal{M}_0$ is convex. To address this fact, we consider a convex relaxation $\mathcal{M}_1 = \{\mathbf{w} \in [0, 1]^n \mid \|\mathbf{w}\|_1 = h\}$ that leads to the optimization problem as a linear integer problem. Therefore, the Equation (3.1) reduces to a discrete minimization problem

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{M}_1} F(\mathbf{w}). \quad (3.2) \quad \boxed{\text{eq:relaxed}}$$

Since $\mathcal{M}_1$ is convex and compact and F is concave (cf. **Theorem 3.1** below) then the local minima of F over $\mathcal{M}_1$ will attain in $\mathcal{M}_0 \subset \mathcal{M}_1$. Now, we will first show that the function F is concave over the domain $\mathcal{M}_1$ and then it's extreme value occurs on the boundary $\partial \mathcal{M}_1$.

Theorem 3.1. *The function F defined in Equation (3.1) is twice continuously differentiable and concave on $\mathcal{M}_1$. Moreover, F is superharmonic, i.e., $\Delta F(\mathbf{w}) \leq 0$, and its Hessian satisfies the uniform bound:*

$$\|\nabla^2 F(\mathbf{w})\| \leq \frac{2KR}{\lambda_0} \quad \text{for all } \mathbf{w} \in \mathcal{M}_1,$$

where

$$K^2 := \sup_{\mathbf{w} \in \mathcal{M}_0} SSY_{\mathbf{w}}, \quad \text{and} \quad \lambda_0 := \min_{\mathbf{w} \in \mathcal{M}_0} \min_{1 \leq i, j \leq n} \lambda_{\min} \left(\sum_{k \neq \{i, j\}}^n w_k \mathbf{x}_k \mathbf{x}_k^\top \right), \quad (3.3) \quad \text{eq:lambda0_def}$$

$$R := \max_{1 \leq i \leq n} \|\mathbf{x}_i\|, \quad \text{and} \quad \varepsilon_{\max} := \max_{1 \leq i \leq n} \sup_{\mathbf{w} \in \mathcal{M}_0} |\varepsilon_i^{\mathbf{w}}|. \quad (3.4) \quad \text{eq:epsmax_def}$$

Proof. We first establish the superharmonic property of F then the concavity of F , finally prove the bound for the Hessian.

First-order and Second-order derivatives: The first derivative of function F in (3.1) with respect to w_i is:

$$\frac{\partial}{\partial w_i} F(\mathbf{w}) = (y_i - (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_i)^2. \quad (3.5) \quad \text{eq:first_deriv}$$

Now, from the Proposition A.1, the estimating equation is $\sum_{i=1}^n w_i (y_i - (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_i) \mathbf{x}_i = 0$, Applying the implicit function theorem:

$$\begin{aligned} 0 &= \frac{\partial}{\partial w_i} \sum_{j=1}^n w_j (y_j - (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_j) \mathbf{x}_j \\ &= (y_i - (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_i) \mathbf{x}_i + \sum_{j \neq i}^n w_j \left(\frac{\partial}{\partial w_i} (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_j \right) \mathbf{x}_j \\ &= \varepsilon_i^{\mathbf{w}} \mathbf{x}_i - \left(\sum_{j \neq i} w_j \mathbf{x}_j \mathbf{x}_j^\top \right) \frac{\partial}{\partial w_i} \hat{\beta}^{\mathbf{w}} \\ \therefore \frac{\partial}{\partial w_i} \hat{\beta}^{\mathbf{w}} &= \left(\sum_{j \neq i} w_j \mathbf{x}_j \mathbf{x}_j^\top \right)^{-1} \varepsilon_i^{\mathbf{w}} \mathbf{x}_i \end{aligned} \quad (3.6) \quad \text{eq:beta_deriv}$$

where $\varepsilon_i^{\mathbf{w}} := y_i - (\hat{\beta}^{\mathbf{w}})^\top \mathbf{x}_i$. Also the second-order derivative follows as:

$$\frac{\partial^2}{\partial w_j \partial w_i} F(\mathbf{w}) = -2 \varepsilon_i^{\mathbf{w}} \varepsilon_j^{\mathbf{w}} (\mathbf{x}_i^\top \left(\sum_{k \neq \{i, j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top \right)^{-1} \mathbf{x}_j). \quad (3.7) \quad \text{eq:second_deriv}$$

Superharmonicity: The Laplacian of the function F : $\Delta F(\mathbf{w}) = \sum_{i=1}^n \frac{\partial^2}{\partial w_i^2} F(\mathbf{w})$. Using (3.5) and (3.6), we compute:

$$\begin{aligned} \Delta F(\mathbf{w}) &= -2 \sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2 (\mathbf{x}_i^\top \left(\sum_{j \neq i} w_j \mathbf{x}_j \mathbf{x}_j^\top \right)^{-1} \mathbf{x}_i) \\ &= -2 \sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2 d^2 \left(0, \mathbf{x}_i \middle| \sum_{j \neq i} w_j \mathbf{x}_j \mathbf{x}_j^\top \right) \leq 0. \end{aligned}$$

Recognizing the term $d^2\left(0, \mathbf{x}_i \left| \sum_{j \neq i}^h w_j \mathbf{x}_j \mathbf{x}_j^\top \right. \right) = \mathbf{x}_i^\top \left(\sum_{j \neq i}^h w_j \mathbf{x}_j \mathbf{x}_j^\top \right)^{-1} \mathbf{x}_i$ as the squared Mahalanobis distance, which is non-negative and we conclude $\Delta F(\mathbf{w}) \leq 0$. Hence, F is superharmonic.
Concavity: Let $\boldsymbol{\eta} \in \mathbb{R}^n$ with $\|\boldsymbol{\eta}\| = 1$. Then by using (3.7) we found:

$$\langle \nabla^2 F(\mathbf{w}) \boldsymbol{\eta}, \boldsymbol{\eta} \rangle = -2 \sum_{i,j} \varepsilon_i^{\mathbf{w}} \varepsilon_j^{\mathbf{w}} \eta_i \eta_j H_{ij}(\mathbf{w}), \quad (3.8) \quad \text{eq:hessian_inn}$$

with the scalar $H_{ij}(\mathbf{w}) := \mathbf{x}_i^\top A_{ij} \mathbf{x}_j$, and the matrix $A_{ij} := \left(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top \right)^{-1}$.

Define $\min_{1 \leq i,j \leq n} \lambda_{\min}^{ij}(\mathbf{w}) := \lambda_{\min} \left(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top \right)$. Here, by Equation (3.3) λ_0 be the minimum eigenvalue of the symmetric positive definite matrix $\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top$. Then $A_{ij} \preceq \frac{1}{\lambda_0} I$, and hence:

$$H_{ij}(\mathbf{w}) \leq \frac{1}{\lambda_0} \|\mathbf{x}_i\| \|\mathbf{x}_j\|. \quad (3.9) \quad \text{eq:H_bound}$$

Substituting (3.9) into (3.8):

$$\begin{aligned} \langle \nabla^2 F(\mathbf{w}) \boldsymbol{\eta}, \boldsymbol{\eta} \rangle &\leq -\frac{2}{\lambda_0} \sum_{i,j} |\varepsilon_i^{\mathbf{w}} \eta_i| \|\mathbf{x}_i\| \cdot |\varepsilon_j^{\mathbf{w}} \eta_j| \|\mathbf{x}_j\| \\ &= -\frac{2}{\lambda_0} \left(\sum_{i=1}^n |\varepsilon_i^{\mathbf{w}} \eta_i| \|\mathbf{x}_i\| \right)^2 \leq 0. \end{aligned}$$

This shows that $\nabla^2 F(\mathbf{w})$ is negative semi-definite, so F is concave in $\mathcal{M}_0$.

Hessian bound: From (3.8) and (3.9), we have:

$$\sup_{\|\boldsymbol{\eta}\|=1} |\langle \nabla^2 F(\mathbf{w}) \boldsymbol{\eta}, \boldsymbol{\eta} \rangle| \leq \frac{2}{\lambda_0} \left(\sum_{i=1}^n |\varepsilon_i^{\mathbf{w}} \eta_i| \|\mathbf{x}_i\| \right)^2. \quad (3.10) \quad \text{eq:hessian_norm}$$

By applying the Cauchy–Schwarz inequality in Equation (3.10) then $\sum_{i=1}^n |\varepsilon_i^{\mathbf{w}} \eta_i| \|\mathbf{x}_i\|$ becomes:

$$\sum_{i=1}^n |\varepsilon_i^{\mathbf{w}} \eta_i| \|\mathbf{x}_i\| \leq \left(\sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2 \right)^{1/2} \left(\sum_{i=1}^n \eta_i^2 \|\mathbf{x}_i\|^2 \right)^{1/2}. \quad (3.11) \quad \text{eq:cauchy_schw}$$

Since $\|\boldsymbol{\eta}\| = 1$ and $\|\mathbf{x}_i\| \leq R$ (from (3.4)), we have: $\sum_{i=1}^n \eta_i^2 \|\mathbf{x}_i\|^2 \leq R^2$.

Now, for each $\mathbf{w} \in \mathcal{M}_1$, let $\hat{y}_i^{\mathbf{w}}$ denote the fitted value and the mean of the fitted values $\bar{Y}^{\mathbf{w}} := \frac{1}{n} \sum_{i=1}^n \hat{y}_i^{\mathbf{w}}$, the explained sum of squares $SSE_{\mathbf{w}} := \sum_{i=1}^n (\hat{y}_i^{\mathbf{w}} - \bar{Y}^{\mathbf{w}})^2$, the residual sum of squares $SSR_{\mathbf{w}} := \sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2$, and the total sum of squares $SSY_{\mathbf{w}} := \sum_{i=1}^n (y_i - \bar{Y}^{\mathbf{w}})^2$. These quantities satisfy the decomposition $SSY_{\mathbf{w}} = SSR_{\mathbf{w}} + SSE_{\mathbf{w}} + 2 \sum_{i=1}^n (\hat{y}_i^{\mathbf{w}} - \bar{Y}^{\mathbf{w}}) \varepsilon_i^{\mathbf{w}}$, where the last term is the cross term. If the cross term vanishes (Wooldridge, 2019) (e.g., under ordinary least squares with an intercept), then $SSR_{\mathbf{w}} \leq SSY_{\mathbf{w}}$. It follows that $\sup_{\mathbf{w} \in \mathcal{M}_0} \sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2 \leq SSY_{\mathbf{w}}$.

So let $\max_{1 \leq i \leq n} \sup_{\mathbf{w} \in \mathcal{M}_0} \sum_{i=1}^n (\varepsilon_i^{\mathbf{w}})^2 := n\varepsilon_{\max}^2$ (by Equation (3.4)). Assuming there exists a finite constant $K > 0$ by Equation (3.3) we conclude that the worst-case residual magnitude is bounded by

$$n\varepsilon_{\max}^2 \leq K^2. \quad (3.12) \quad \boxed{\text{eq:max_err}}$$

Referring the Equation (3.12), Equation (3.11) and Remark 3.2, then substituting into (3.10):

$$|\langle \nabla^2 F(\mathbf{w}) \boldsymbol{\eta}, \boldsymbol{\eta} \rangle| \leq \frac{2}{\lambda_0} \cdot \sqrt{n\varepsilon_{\max}} \cdot R \leq \frac{2KR}{\lambda_0},$$

which completes the proof. $\square$

$\langle \text{rem:cons} \rangle$

Remark 3.2. For any $\mathbf{w} \in \mathcal{M}_1$ where $\mathcal{M}_1 = \text{conv}(\mathcal{M}_0)$, the matrix $\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top$ is symmetric positive definite and the data matrix $\mathbf{X}$ has full column rank p . This matrix is invertible for any $\mathbf{w} \in \mathcal{M}_1$ and its inverse is also symmetric positive definite.

Let $\lambda_{\min}^{ij} := \min_{1 \leq i, j \leq n} \lambda_{\min}(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)$.

Then,

$$\min_{\mathbf{w} \in \mathcal{M}_1} \min_{1 \leq i, j \leq n} \lambda_{\min}(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top) \geq \min_{\mathbf{w} \in \mathcal{M}_0} \min_{1 \leq i, j \leq n} \lambda_{\min}(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top).$$

This implies,

$$\max_{\mathbf{w} \in \mathcal{M}_1} \min_{1 \leq i, j \leq n} \lambda_{\max}((\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)^{-1}) \leq \frac{1}{\min_{\mathbf{w} \in \mathcal{M}_0} \min_{1 \leq i, j \leq n} \lambda_{\min}(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)}.$$

We can then bound the maximum norm using the concavity of $\lambda_{\min}(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)$. This follows because the minimum eigenvalue function is concave with respect to the weights w

$$\max_{\mathbf{w} \in \mathcal{M}_1} \|(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)^{-1}\| \leq \max_{\mathbf{w} \in \mathcal{M}_0} \|(\sum_{k \neq \{i,j\}} w_k \mathbf{x}_k \mathbf{x}_k^\top)^{-1}\| = \frac{1}{\lambda_{\min}^{ij}} = \frac{1}{\lambda_0} < \infty.$$

Now, We show that the relaxed problem is equivalent to the original problem.

$\langle \text{prop:relaxed} \rangle$

Proposition 3.3. There holds that F is a concave (by Theorem 3.1) function over $\mathcal{M}_1$ and

$$\min_{\mathbf{w} \in \mathcal{M}_1} F(\mathbf{w}) = \min_{\mathbf{w} \in \mathcal{M}_0} F(\mathbf{w}).$$

Proof. It is direct to see from the definition of L in Equation (2.1) that $L(\mathbf{w}, \boldsymbol{\beta})$ is a linear function of $\mathbf{w}$ for any fixed $\boldsymbol{\beta}$. Therefore, the minimum $F(\mathbf{w})$ of linear functions $L(\mathbf{w}, \boldsymbol{\beta})$ is a concave function of $\mathbf{w}$.

For a concave function, the minimum over a convex set is achieved at the extreme points of the set (by Theorem 3.1). Since $\mathcal{M}_1$ is convex and its extreme points are $\mathcal{M}_0$, we have the desired result. $\square$

We now present the Frank-Wolfe algorithm for solving the minimization problem Equation (3.2). It is a concave minimization problem over a convex set. For a given iterate $\mathbf{w}_{\text{FW}}^{(k)}$ at current iteration $k \in \mathbb{N}$, we first find a feasible direction $\mathbf{d}^{(k)}$ through minimizing a linear approximation of F :

$$\mathbf{d}^{(k)} = \operatorname{argmin}_{\mathbf{d} \in \mathcal{M}_1} \left\langle \nabla F(\mathbf{w}_{\text{FW}}^{(k)}), \mathbf{d} \right\rangle, \quad (3.13) \quad \boxed{\text{eq:dk}}$$

where $\nabla F(\mathbf{w}_{\text{FW}}^{(k)})$ is the gradient of F at $\mathbf{w}_{\text{FW}}^{(k)}$. We then update the solution by

$$\mathbf{w}_{\text{FW}}^{(k+1)} = \mathbf{w}_{\text{FW}}^{(k)} + \alpha^{(k)}(\mathbf{d}^{(k)} - \mathbf{w}_{\text{FW}}^{(k)}), \quad (3.14) \quad \boxed{\text{eq:fw_update}}$$

where $\alpha^{(k)} \in [0, 1]$ is obtained from a line search on the interval $[0, 1]$:

$$\alpha^{(k)} = \arg \min_{\alpha \in [0, 1]} F(\mathbf{w}_{\text{FW}}^{(k)} + \alpha(\mathbf{d}^{(k)} - \mathbf{w}_{\text{FW}}^{(k)})).$$

Algorithm 2 F-W algorithm (Lacoste-Julien, 2016) for C-step

$\langle \text{alg:FW} \rangle$

- 1: Let $\mathbf{w}_{\text{FW}}^{(0)} \in \mathcal{M}_0$
- 2: **for** $k = 0, 1, \dots, K$ **do**
- 3: Compute $\mathbf{d}^{(k)} := \arg \min_{\mathbf{d} \in \mathcal{M}_0} \left\langle \nabla F(\mathbf{w}_{\text{FW}}^{(k)}), \mathbf{d} \right\rangle$
- 4: Let $\mathbf{s}^{(k)} = \mathbf{d}^{(k)} - \mathbf{w}_{\text{FW}}^{(k)}$
- 5: Compute $g_k := \left\langle -\nabla F(\mathbf{w}_{\text{FW}}^{(k)}), \mathbf{s}^{(k)} \right\rangle$
- 6: **if** $g_k \leq \varepsilon$ **then**
- 7: **return** $\mathbf{w}_{\text{FW}}^{(k)}$
- 8: **end if**
- 9: Line search: $\alpha^{(k)} := \arg \min_{\alpha \in [0, 1]} F(\mathbf{w}_{\text{FW}}^{(k)} + \alpha \mathbf{s}^{(k)})$
- 10: Update: $\mathbf{w}_{\text{FW}}^{(k+1)} = \mathbf{w}_{\text{FW}}^{(k)} + \alpha^{(k)} \mathbf{s}^{(k)}$
- 11:
- 12: **if** F is concave **then**
- 13: $\alpha^{(k)} = 1$
- 14: **end if**
- 15: **end for**
- 16: **return** $\mathbf{w}_{\text{FW}}^{(K)}$

We next show that the C-step method is equivalent to the Frank-Wolfe algorithm.

$\langle \text{prop:fw_cstep} \rangle$ **Proposition 3.4.** *When starting from the same iterate $\mathbf{w}_{\text{FW}}^{(k)} = \mathbf{w}_{\text{C}}^{(k)}$, the Frank-Wolfe algorithm and the C-step method generate the same $\mathbf{w}$ in the next iteration:*

$$\mathbf{w}_{\text{FW}}^{(k+1)} = \mathbf{w}_{\text{C}}^{(k+1)}.$$

Proof. We first compute the feasible direction $\mathbf{d}^{(k)}$ Equation (3.13) in the Frank-Wolfe algorithm. To this end, we need to compute the gradient $\nabla F(\mathbf{w}_{\text{FW}}^{(k)})$ of F at $\mathbf{w}_{\text{FW}}^{(k)}$. By the definition of F in Equation (3.1),

$$\frac{\partial F(\mathbf{w})}{\partial w_j} = r_j(\boldsymbol{\beta}^{\mathbf{w}}) + \sum_{i=1}^n w_i \frac{\partial r_i(\boldsymbol{\beta}^{\mathbf{w}})}{\partial w_j} = r_j(\boldsymbol{\beta}^{\mathbf{w}}) + \sum_{i=1}^n w_i \nabla_{\boldsymbol{\beta}}^{\top} r_i(\boldsymbol{\beta}^{\mathbf{w}}) \frac{\partial \boldsymbol{\beta}^{\mathbf{w}}}{\partial w_j}.$$

Note that $L(\mathbf{w}, \boldsymbol{\beta})$ is a quadratic function of $\boldsymbol{\beta}$ for any fixed $\mathbf{w}$. Its minimizer $\boldsymbol{\beta}^{\mathbf{w}}$ could be obtained from the [Proposition A.1](#). Substituting $\boldsymbol{\beta}^{\mathbf{w}}$ into the partial derivative of F , we have

$$\frac{\partial F(\mathbf{w})}{\partial w_j} = r_j(\boldsymbol{\beta}^{\mathbf{w}}).$$

It follows from Equation [\(3.13\)](#) that

$$\mathbf{d}^{(k)} = \arg \min_{\mathbf{d} \in \mathcal{M}_1} \sum_{j=1}^n d_j r_j(\boldsymbol{\beta}^{\mathbf{w}_{\text{FW}}^{(k)}}) = \arg \min_{\mathbf{d} \in \mathcal{M}_0} \sum_{j=1}^n d_j r_j(\boldsymbol{\beta}^{\mathbf{w}_{\text{FW}}^{(k)}}).$$

This is exactly the $\mathbf{w}$ -update step in the C-step method Equation [\(2.2\)](#) when starting from $\mathbf{w}_{\text{FW}}^{(k)} = \mathbf{w}_C^{(k)}$. It implies that

$$\mathbf{d}^{(k)} = \mathbf{w}_C^{(k+1)}.$$

We next compute the step size $\alpha^{(k)}$ in Equation [\(3.14\)](#). By [Proposition 3.3](#), we know that F is a concave function. It follows from [\(Yurtsever and Sra, 2022, Proposition 3.1\)](#) that $\alpha^{(k)} = 1$ and

$$\mathbf{w}^{(k+1)} = \mathbf{d}^{(k)} = \mathbf{w}_C^{(k+1)}.$$

Consequently, the Frank-Wolfe algorithm reduces to the C-step method. $\square$

Since the C-step method is equivalent to the Frank-Wolfe algorithm, we will discard the superscript “FW” and “C” and use $\mathbf{w}^{(k)}$ to denote the iterate at current iteration k in the rest of the paper.

4 Convergence Analysis

ec:convergence We will analyze the convergence of the C-step method in this section. From the results in [Section 3](#), the C-step method could be reformulated as the Frank-Wolfe algorithm. The analysis tools for the Frank-Wolfe algorithm could then be employed to analyze the convergence of the C-step method. However, most of the existing convergence analysis for the Frank-Wolfe algorithm are derived for convex objective functions. That is, they could not be applied directly to the concave objective function F in Equation [\(3.1\)](#). We will generalize the existing convergence analysis for the Frank-Wolfe algorithm to concave objective functions in this section.

We first review some notions for the Frank-Wolfe algorithm. The *Frank-Wolfe (FW) gap* of F at $\mathbf{w}^{(k)}$ is defined as follows:

$$g_k = \max_{\mathbf{d} \in \mathcal{M}_1} \left\langle \mathbf{d} - \mathbf{w}^{(k)}, -\nabla F(\mathbf{w}^{(k)}) \right\rangle.$$

It is a standard quantity in the convergence analysis for the Frank-Wolfe algorithm. It is always nonnegative and we say a point $\mathbf{w}^{(k)}$ is a *stationary point* of F if $g_k = 0$. When the objective function is convex, the Frank-Wolfe gap dominates the primal gap $\|\mathbf{w}^{(k)} - \mathbf{w}^*\|$, where $\mathbf{w}^*$ is the optimal solution. For non-convex objective functions, we will focus on the analysis of the Frank-Wolfe gap.

We next present the convergence analysis for the C-step method.

fw_convergence **Theorem 4.1.** Let $\mathbf{w}^{(k)}$ be the iterate generated by the C-step method. The running minimal FW gap $\tilde{g}_T := \min_{0 \leq k \leq T-1} g_k$ satisfies that

$$\tilde{g}_T \leq \frac{F(\mathbf{w}^{(0)}) - \min_{\mathbf{w} \in \mathcal{M}_1} F(\mathbf{w})}{T}, \quad T \in \mathbb{N}.$$

Proof. Since F is concave, the result follows immediately from (Yurtsever and Sra, 2022, Lemma 2.1) showing the non-asymptotic convergence of the Frank-Wolfe method on concave functions. $\square$

We remark that the objective function F is concave and the C-step might converge to a local minimum of F . That is, the primal gap $\|\mathbf{w}^{(k)} - \mathbf{w}^*\|$ might not converge to zero. In Lacoste-Julien (2016), the convergence rate $\mathcal{O}(\frac{1}{\sqrt{T}})$ of the FW gap is derived for general non-convex functions. We could derive a better convergence rate $\mathcal{O}(\frac{1}{T})$ due to the concavity of F .

5 Simulation Study

sec:sim We illustrate the findings of Section 4 through a simulation study. A primary numerical difficulty in evaluating the convergence rate of Algorithm 2 lies in the tendency of the iterations to become stationary. Since our theoretical results are non-asymptotic, the approach is to examine the numerical convergence within the preasymptotic regime. For a given dataset and initialization seeds, this regime may be too brief to allow meaningful statistical analysis. Consequently, we focus instead on the worst-case performance computed over randomly chosen seeds—either for a fixed dataset or across multiple datasets drawn from a given population. Denoting by g_t^{WC} the worst-case FW gap at iteration t , the figure below shows that it typically decreases monotonically within the preasymptotic regime. Hence, we analyze this regime rather than the running minimum asymptotic gap $\tilde{g}_t$ used in Theorem 4.1. In practice, the convergence rate of g_t^{WC} is estimated via ordinary least squares applied to the linear regression model

$$\log(g_t^{\text{WC}}) = \beta_0 + \beta_1 \log(t) + \varepsilon \quad \text{assuming i.i.d. normal errors } \varepsilon. \quad (5.1) \quad \boxed{\text{eq:slope}}$$

The preasymptotic regime was defined as the lower two-thirds of the $\log(t)$ range. Theoretical expectations were set as $\beta_1 = -1$ and $\beta_0 = (n-1)\text{tr}(\Sigma)$, with the worst-case total variance estimated empirically. In your simulation, β_0 is the empirical intercept obtained from the robust regression of $\log(g_t^{\text{WC}})$ on $\log(t)$. All implementations were done in pure Python and executed in CPU mode on a 64-bit operating system.

Synthetic data: In our synthetic data experiments, we generate regression datasets that include bad leverage points—outliers that simultaneously distort predictors and responses—to evaluate the convergence behavior and robustness of the LTS (Least Trimmed Squares) regression algorithm implemented in our study. The data are produced using the a mixture of clean and contaminated observations. The majority (clean) data are drawn from a linear model followed from Rousseeuw and Driessen (2006) and a fraction of the data, determined by the contamination level is replaced by outlier observations with predictors shifted far from the main cluster,

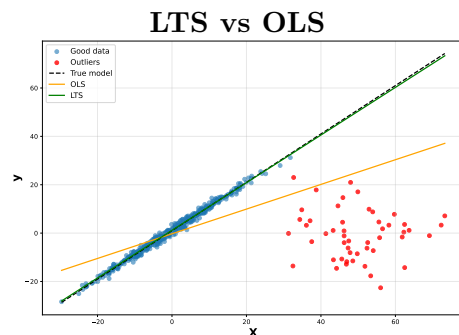


Figure 1: Synthetic data

and responses replaced with independent random noise thus breaking the linear relationship. The clean and contaminated samples are combined and randomly shuffled to form the final dataset.

We systematically vary the sample size $n = 500, 1000$, dimension $d = 5, 10$ including intercept), and contamination level $\varepsilon = 0.1, 0.15$ to examine their effect on convergence. For each configuration, multiple random seeds are used to capture the worst-case convergence behavior of the LTS C-step algorithm Initialization follows the warm-start strategy implemented in `warmstart`, where random subsets of size $p + 1$ (with $p = d$) are used to compute initial least-squares estimates. The trimming parameter is set as $h = \left\lfloor \frac{n+p+1}{2} \right\rfloor$, and the algorithm iteratively refines the estimate through repeated C-steps until convergence or a maximum iteration limit is reached. **Figure 1** shows the regression result (for small observation) and C-step convergence using our discovered way.

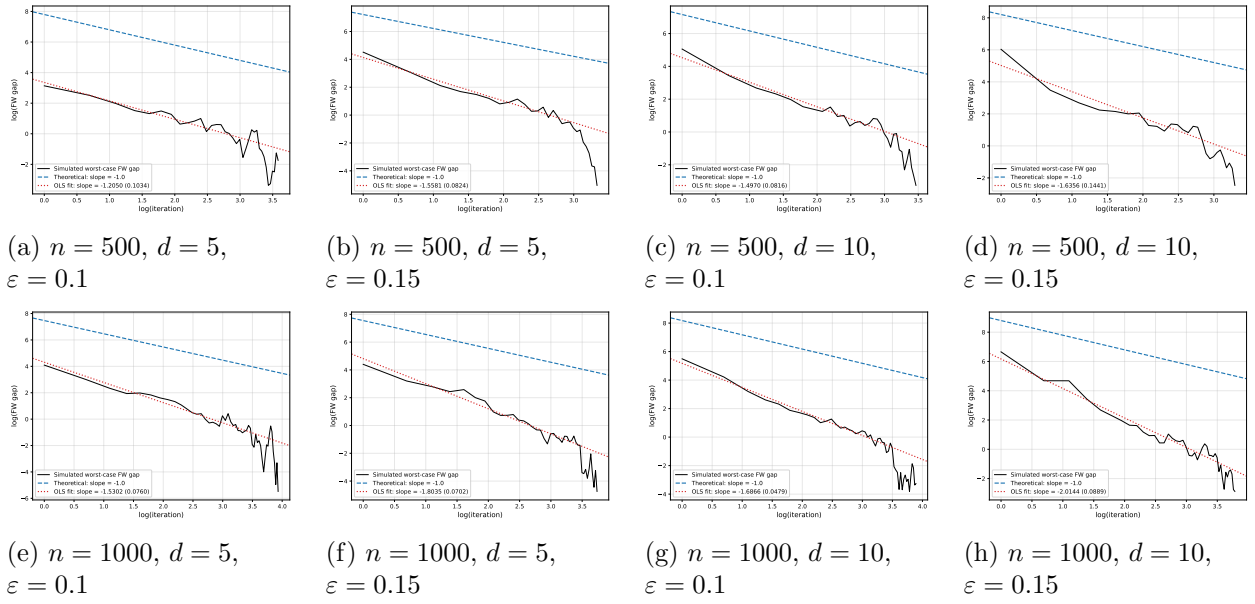


Figure 2: worst-case FW gap for variable n, d , and ε .

To assess convergence rates, we record the Frank–Wolfe–type gap over iterations and estimate the empirical slope of Equation (5.1). The preasymptotic regime—defined as the lower 2/3 of the $\log(t)$ range—is used to estimate β_1 , corresponding to the empirical convergence rate. Computations are performed in plane Python (NumPy + SciPy) and executed on a standard 64-bit CPU environment. Typical runs consist of 200 replications with up to 300 C-step iterations each. Runtime per configuration is on the order of a few minutes, with the largest datasets ($n = 1000, d = 10$) being the most computationally demanding due to repeated subset fitting. Overall, the simulations confirm that higher contamination levels and dimensionality slow convergence and can obscure the theoretical rate of -1 , while robust initialization and LTS-based slope estimation yield stable and interpretable convergence patterns even under severe contamination.

Figure 2 displays the worst-case (i.e., maximum over all replications) FW gap (and running minimum FW gap plots. Applying an upper-sided Student’s t -test of $H_0 : \beta_1 = -1$ vs $H_1 : \beta_1 > -1$ across all scenarios considered, the p -values never dropped below $1 - (1 - 0.05)^{1/36} \approx 0.0014$, which is the Šidák-adjusted limit to maintain a family-wise 0.05 test size, the composite alternative hypothesis was rejected corroborating that the worst-case pre-asymptotic rate was never slower

than $\mathcal{O}(1/t)$.

California Housing Data: Consider the Housing dataset obtained from the Kaggle notebook [ShtraussLearning \(2025\)](#), which contains various numerical features describing residential properties and a continuous target variable representing house prices. After removing missing values (1.25%), selecting numerical predictors, and standardizing all features, the training set consists of approximately $n \approx 16.5\text{k}$ observations with $d = 9$ continuous variables. To assess the robustness and convergence behavior of the Least Trimmed Squares (LTS) regression algorithm under contamination, artificial bad leverage points.

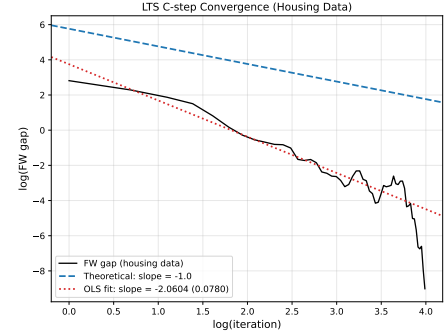


Figure 3: Housing data (worst-case FW gap)

For each contamination level, a Monte Carlo simulation with 100 replications was performed, where each replication applied a warm-start initialization (followed by convergence to 200 C-step iterations). The objective function (sum of the h smallest squared residuals) was recorded at each iteration, and the maximum Frank–Wolfe style gap across replications was computed to evaluate convergence. A robust regression of $\log(g_t)$ on $\log(t)$, where g_t denotes the gap at iteration t , yielded an estimated slope $\hat{\beta}_1 \approx -1.8035$ with standard error $\text{se}(\hat{\beta}_1) \approx 0.0702$. Based on a two-sided Student’s t -test at significance level $\alpha = 0.05$, the null hypothesis $\beta_1 = -1$ was not rejected, corroborating an $\mathcal{O}(1/t)$ worst-case (Figure 3) convergence rate of the LTS C-step algorithm on this dataset, even in the presence of moderate contamination.

6 Summary and Discussion

Discussion. We reiterate and emphasize the importance of non-asymptotic convergence guarantees for LTS – or any other machine learning algorithm for that matter – that hold in a uniform fashion with respect to the sample size, space dimension, hyperparameters, initial seeding or optimization “intricacies” like the curvature constant or geometric aspects of the domain. Building upon the recent precedent ([Yurtsever and Sra, 2022](#)) in the context of convex-concave procedure (CCCP), we proved that LTS algorithm is a special case of FW. In doing so, we not only were able to perfectly carry over their results to LTS but developed an FW variant for general non-smooth concave objectives and proved an $\mathcal{O}(1/t)$ convergence rate.

Extensions. This work can be extended in several meaningful directions. One promising avenue is the integration of advanced regression techniques such as Ridge Regression, LASSO, and Principal Component Regression (PCR). These methods introduce different regularization mechanisms and dimensionality reduction strategies particularly in high-dimensional or noisy environments. A systematic comparison of the proposed method with these approaches could reveal new insights into their convergence behavior, and predictive performance in robust statistics. Furthermore, exploring adaptive that combine these techniques may lead to more efficient and generalizable optimization algorithms for large-scale learning problems.

A Auxiliary Results

A.1 OLS Estimator

Proposition A.1. For fixed weights $\mathbf{w} \in \mathcal{M}_1$, the unique minimizer $\beta^{\mathbf{w}}$ of the function in Equation (2.10) takes an explicit form of

$$\hat{\beta}^{\mathbf{w}} = \left(\sum_{i=1}^n w_i \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\sum_{i=1}^n w_i y_i \mathbf{x}_i^\top \right).$$

Proof. Here, we assume any $\lfloor (n+p+1)/2 \rfloor$ rows of the data matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n]^\top$ have full column rank p . This ensures that the matrix of second moments $\sum_{i \in I^{(k)}} \mathbf{x}_i \mathbf{x}_i^\top$ is positive definite for any $I^{(k)}$ with cardinality $h \geq \lfloor (n+p+1)/2 \rfloor$. Now, taking the gradient of function $L(\mathbf{w}, \beta)$ with respect to β , $\nabla_{\beta} L(\mathbf{w}, \beta) = \sum_{i=1}^n w_i (y_i - \beta^\top \mathbf{x}_i) \mathbf{x}_i$. Therefore, minimization over β is uniquely solved by using Fermat's optimality conditions for any fixed $\mathbf{w} \in \mathcal{M}_1$. Now, the estimation equation is:

$$\begin{aligned} 0 &= \nabla_{\beta} L(\mathbf{w}, \beta) \\ &= \sum_{i=1}^n w_i (y_i - \beta^\top \mathbf{x}_i) \mathbf{x}_i \\ &= \sum_{i=1}^n w_i y_i \mathbf{x}_i^\top - \left(\sum_{i=1}^n w_i \mathbf{x}_i \mathbf{x}_i^\top \right) \beta \end{aligned}$$

So the solution for $\hat{\beta}^{\mathbf{w}}$ of Equation (2.1) is $\hat{\beta}^{\mathbf{w}} = \left(\sum_{i=1}^n w_i \mathbf{x}_i \mathbf{x}_i^\top \right)^{-1} \left(\sum_{i=1}^n w_i y_i \mathbf{x}_i^\top \right)$ $\square$

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